$\bar{\nu}_e$ Appearance Analysis with the 3-Flavour Joint Fit Framework for T2K Run 1-9c ($1.493\,797\,7 \times 10^{21}$-POT Neutrino and $1.123\,559 \times 10^{21}$-POT Anti-Neutrino) Data Sets

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Abstract

[Caveat (2018/04/03): Currently the sensitivity plots and results are shown without the weight of the 4 control samples. These un-weighted sensitivities are a place-holder until the weighted analysis has finished running (expected in a few days)]

We report the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour, 5 sample $\nu/\bar{\nu}$ analysis framework, performed by the *VALOR* group, on the combined Run 1-9c data set, using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam and is an update to [13] The joint fit framework is used, whose details can be found in [14]. A joint analysis of all the Super-K samples, in both neutrino and antineutrino beam modes, is used in order to constrain the oscillation and systematic parameters, achieving a better sensitivity to $\bar{\nu}_e$ appearance. The Run 1-9c data set corresponds to an integrated J-PARC neutrino beam exposure of $1.493\,797\,7 \times 10^{21}$ POT in neutrino mode and $1.123\,559 \times 10^{21}$ POT in antineutrino mode.

This analysis predicts 268.535 ± 16.387 (syst) for neutrino μ-like, 95.5347 ± 9.7742 (syst) for antineutrino μ-like, 73.7843 ± 8.5898 (syst) for neutrino e-like, 11.75 ± 3.43 (syst) for antineutrino e-like and 6.92422 ± 2.63239 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively for neutrino μ-like, for antineutrino μ-like, for neutrino e-like, for antineutrino e-like and for neutrino ν_eCC1π^+-like events.

A parameter β, that multiplies the $\bar{\nu}_\mu \to \bar{\nu}_e$ probability by either 0 or 1, is introduced in order to obtain the significance to $\bar{\nu}_e$ appearance. All the oscillation parameters, mass hierarchy included, are marginalised. The significance result is shown as a p-value. Both a rate+shape and a rate-only

analysis are performed.

For the Run 1-9c set the number of observed events in the antineutrino mode e-like spectrum is and the measured rate+shape test-statistic, $\Delta\chi^2$, . Taking no $\bar{\nu}_e$ appearance as our null hypothesis we find a p-value of ±for the rate-only analysis and a p-value of ±for the rate+shape analysis. Taking $\bar{\nu}_e$ appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of ±for the rate-only analysis and a p-value of ±for the rate+shape analysis.

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Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	gaussian	0.0857 ± 0.0046
$\sin^2 2\theta_{12}$	gaussian	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	uniform	$[2.3; 2.7] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	gaussian	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	uniform	$[-\pi; +\pi]$
Mass Hierarchy	uniform	0.5 normal, 0.5 inverted

Table 1: Treatment of the oscillation parameters in the $\bar{\nu}_e$ appearance analysis. All the gaussian priors are from [83]. The parameter $\sin^2 2\theta_{13}$ is constrained with the measurement of reactor experiments using the gaussian prior. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [83], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [22].

1. Introduction

In this technical note, we report the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the VALOR group, on the combined Run 1-9c data set, using data taken in both neutrino mode ($1.493\,797\,7 \times 10^{21}$), with the polarity set to forward horn current (FHC), and antineutrino ($1.123\,559 \times 10^{21}$) mode with polarity set to reverse horn current (RHC). The same procedure used in the 2016 $\bar{\nu}_e$ appearance analysis [13] is used, whilst working in the analysis framework outlined in [14]. A parameter β, that multiplies by either 0 or 1 the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation probability, is used in order to obtain the significance to $\bar{\nu}_e$ appearance. All of the oscillation and systematic parameters, mass hierarchy included, are marginalised. The priors used to marginalise the oscillation parameters can be found in Tab 1.

The significance to $\bar{\nu}_e$ appearance is given as a p-value. Two different test statistics are used. The total number of observed events (N_{obs}) in the antineutrino mode e-like sample, without using any information from the reconstructed energy shape, is used to calculate the rate-only p-value. $\Delta\chi^2 = \chi^2(\beta = 0) - \chi^2(\beta = 1)$ using both rate and shape information from the five available samples is used to calculate the rate+shape p-value.

1.1. Improvements to the Previous Analysis

There have been a number of improvements since this analysis was previously performed, in 2016 [13]:

- The FHC exposure has increased by 113 % from 6.914×10^{20} to $1.493\,797\,7 \times 10^{21}$
- The RHC exposure has increased by 50 % from 4.011×10^{20} to $1.123\,559 \times 10^{21}$
- An additional Super-K event sample has been added, neutrino mode multi-ring electron-like
- The Super-K event reconstruction algorithm has changed from using APFit to FitQun [19]
- New cross-section model parameters are implemented, as detailed in [14]

1.2. Construction of Nominal Super-K Monte Carlo Templates

The analysis templates used in this analysis were constructed in an analogous way to the procedure used for the 3-flavour joint analysis in [14]. The parameter of interest β which reweights the $\bar{\nu}_e$ appearance signal component is kept fixed to either 0 (corresponding to no appearance) or 1 (corresponding to appearance consistent with the current best knowledge of the PMNS matrix given by the simultaneous T2K+reactor Run 1-8 joint fit result) and is applied to all the Super-K samples, though its impact is negligible in all the samples except the antineutrino mode e-like sample.

In this analysis a comparison of the reconstructed energy (E_r) and reconstructed lepton angle (θ) spectrum of single ring events, including the ν_e CC1π^+-like sample, observed at Super-K with the predicted spectrum is used to build the test statistic to calculate the significance to $\bar{\nu}_e$ appearance. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in previous analyses, and a single θ bin is used. The predicted number $N_{SK;r,\theta}$ of RHC single-ring e-like events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} \beta \cdot P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \tag{1}$$

In equation 1, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and reconstructed lepton angle bin θ. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on a vector of nuisance (systematic) parameters $\vec{f}$, which is function of the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, and the reconstructed lepton angle bin θ. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; This is exactly how the analysis templates are built in the joint fit analysis, with the only difference that the $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ component can be switched on or off by the parameter β in each Super-K sample. In Fig **??** the predicted spectra are compared with the observed data. Also the 2-dimensional spectrum of the $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ component, that can be switched on or off by the parameter β is shown.

1.3. Calculation of the Marginal Likelihood

In the analyses presented here both the likelihood and the marginal likelihood are used. First the log-likelihood function is calculated as follows:

$$-2 \ln \lambda(\beta, \mathbf{a}) = 2 \cdot \sum_{i=0}^{N-1} \left(n_i^{obs} \cdot \ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp} - n_i^{obs}) \right) \tag{2}$$

where n_i^{obs} is the observed number of events in the ith bin and $n_i^{exp} = n_i^{exp}(\beta; \mathbf{a})$ is the corresponding expected number of events, N is the number of reconstructed energy and lepton angle bins and $\mathbf{a}$ is a vector of systematic parameters.

The total likelihood is then given by the sum of $-2 \ln \lambda(\beta, \mathbf{a})$ from all of the Super-K samples

$$-2 \ln \lambda(\beta, \mathbf{a}) = \sum_{s=1}^{N_{samples}} -2 \ln \lambda_s(\beta, \mathbf{a}) \tag{3}$$

where $N_{samples}$ corresponds to the number of Super-K samples used in the analysis. The marginal likelihood is then computed by considering an ensemble of MC predictions, generated based on priors on nuisance parameters, as in [14].

$$\chi^2_{marg}(\beta) = -2 \ln \lambda(\beta) = -2 \ln \left(\frac{1}{n} \sum_{i=1}^{n} \lambda(\beta, \mathbf{a}) \right) \tag{4}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the n throws, not a weighted average of the log-likelihood.

2. $\bar{\nu}_e$ Appearance Methodology

The significance of $\bar{\nu}_e$ appearance is given as a p-value by introducing the parameter β in order to compare the data to two null hypotheses: a) No $\bar{\nu}_e$ appearance and b) $\bar{\nu}_e$ appearance according to the current best knowledge of the PMNS matrix. Two different test statistics are used: rate-only and rate+shape.

An ensemble of toys are generated for each null hypothesis, randomising nuisance parameters according to the priors in Tab 1. In order to constrain the oscillation and systematic model parameters using T2K data from the "control samples" (neutrino mode single-ring and multi-ring e-like and both neutrino and antineutrino mode single-ring μ-like), the toy distributions are weighted according to the following method, which ensures that the test-statistic distributions are filled with only the most likely toys and maintains correlations between all nuisance parameters in the 4 "control samples":

1. A large number ($>$ 20k) of toy experiments are produced taking random parameters from the distributions in Tab 1.
2. The 4 sample (i.e. excluding antineutrino mode single-ring e-like) likelihood is calculated for each toy using equation 5. Then the maximum likelihood, λ_{max}, is stored.
3. A toy experiment is produced taking random parameters from the distributions in Tab 1 to form the proposed distribution, $T_{proposed}$.
4. The 4 sample likelihood of $T_{proposed}$ is found: $\lambda_{proposed}$.
5. A random number R is drawn uniformly from $\{0 : 1\}$.
6. If $\frac{\lambda_{proposed}}{\lambda_{max}}$ is greater than R, $T_{proposed}$ is accepted and used in the distribution.
7. The procedure is repeated from step 3 until 20k of $T_{proposed}$ toys have been accepted, denoted $T_{accepted}$.
8. The statistic is calculated according to sections 2.1 and 2.2 for each $T_{accepted}$ to produce test-statistic distributions.

$$\lambda\left(Toy;Data\right) = \prod_{s=1}^{N_{samples}=4} \lambda_s\left(Toy;Data\right) \tag{5}$$

It is worth noting that this method of weighting the toy distributions is mathematically equivalent to performing a likelihood ratio fit of the "control samples" and using the posterior distribution of the nuisance parameters as a prior to extract the systematic parameters for a toy experiment.

At this point, the rate-only and rate+shape analysis methods diverge.

2.1. Rate-only Analysis Method

The total number of observed events, N_{obs}, in the antineutrino mode e-like sample, without using any information from the reconstructed energy and angle distribution, is used to quote the rate-only p-value.

The expected distribution of antineutrino mode e-like events, under statistical and systematic fluctuations, is produced using 20k (throws of nuisance parameters) x 10k (throws of statistical fluctuations) toy data sets to ensure the nuisance parameter space was scanned effectively. In this method, the toy experiments used for the 20k nuisance parameter throws are the weighted toys described in section 2. For each of these toys, i_{toy}, 10k statistical throws are performed using a Poisson distribution with mean equal to the number of antineutrino mode single-ring e-like events in the toy. The result of each statistical throw, N_{exp}, is stored in a rate-only test-statistic distribution for each toy, $f_{itoy}\left(N_{exp}\right)$.

After performing statistical throws, a p-value is calculated for each $f_{itoy}\left(N_{exp}\right)$. For the $\beta = 0$ null hypothesis, a right-tailed p-value is used (see equation 8), while for the $\beta = 1$ null hypothesis, a left-tailed p-value is used (see equation 7). For each null hypothesis, the mean of all p-values is quoted as the result:

$$\text{p-value}_{itoy}(\text{right tail}) = \frac{\int_{N_{obs}}^{+\infty} f_{itoy}\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f_{itoy}\left(N_{exp}\right) dN_{exp}} \tag{6}$$

$$\text{p-value}_{itoy}(\text{left tail}) = \frac{\int_{-\infty}^{N_{obs}} f_{itoy}\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f_{itoy}\left(N_{exp}\right) dN_{exp}} \tag{7}$$

$$\text{p-value} = \frac{1}{n_{toys}} \sum_{itoy=0}^{n_{toys}-1} \text{p-value}_{itoy} \tag{8}$$

As the distribution of the statistic is populated with such a large sample, the statistical uncertainty on the p-value is very small.

2.2. Rate+shape Analysis Method

The rate+shape analysis method uses log-likelihood ratio between the $\beta = 0$ and $\beta = 1$ $\bar{\nu}_e$ appearance null hypotheses as the test-statistic:

$$\Delta\chi^2 = \chi^2_{marg}\left(\beta = 0\right) - \chi^2_{marg}\left(\beta = 1\right) \tag{9}$$

This test statistic simply compares the oscillation with the non-oscillation hypothesis but does not use the best-fit χ^2 value. For this reason the distribution of $\Delta\chi^2$ is expected to take both negative and positive values. It is worth noting that the parameter β is not fit, but is fixed either to 0 or 1. It is also worth noting that marginal χ^2 is calculated by using all 5 Super-K samples.

The $\Delta\chi^2$ distribution is produced by first generating 20k toys produced using the method described in section 2. It is worth noting that this takes into account only systematic fluctuations not statistical fluctuations. The toys are then marginalised using 200k marginalisation toys generated from the priors in Tab 1 and $\chi^2_{marg}\left(\beta\right)$ is calculated according to equation 4. This incorporates the uncertainty on nuisance parameters on the measurement and distributions of the test-statistic.

Once a $\Delta\chi^2$ value is calculated for each toy, it is stored in a single test-statistic distribution, $f\left(\Delta\chi^2\right)$, which is integrated to obtain p-values, for example the right-tailed p-value for the $\beta = 0$ null hypothesis shown in equation 10. Although this is in contrast to the rate-only analysis, where there is one test-statistic distribution per toy, both methods for calculating the p-value are mathematically identical. The method used to calculate the rate-only p-value is used in order to produce distributions of the toy p-value.

$$\text{p-value(right tail)} = \frac{\int_{\Delta\chi^2_{obs}}^{+\infty} f\left(\Delta\chi^2\right) d\Delta\chi^2}{\int_{-\infty}^{+\infty} f\left(\Delta\chi^2\right) d\Delta\chi^2} \tag{10}$$

The number of statistical throws from each toy data set is different between the rate-only and rate+shape analyses. The rate+shape analysis only uses a single statistical fluctuation upon generating each fake-data toy. The reason for this is because the limiting factor in the rate+shape analysis is the calculation of the marginal likelihood. 200k marginalisation toys were used to determine the marginal likelihood in order to explore the whole space of the systematic parameters. More marginalisation toys are used compared to the joint fit analysis, because in this analysis the mass hierarchy, δ_{CP} and all other oscillation parameters are marginalised at the same time whereas they were not in the joint analysis. Furthermore these parameters have a large impact on the $\nu_\mu \to \nu_e$ component which is a significant background in the null hypotheses.

It is worth pointing out that sometimes the rate-only p-value can be smaller than the rate+shape. This can be due to both the discreteness of the rate-only p-value or to the shape information which can discredit the $\bar{\nu}_e$ appearance hypothesis. For this reason it is important to quote both the p-values as our sensitivity.

[Statements about the agreement with P-Theta - waiting for the weighted analysis]

2.3. Statistical Uncertainty on the p-values

The statistical uncertainty on the p-value is calculated assuming that the number of entries in each bin of the expected distribution of the test statistic follow a Poisson distribution. The p-value uncertainty is then derived from the statistical uncertainty of the population. As the p-value is

calculated from the fraction of trials which fall to the right of the data statistic, the p-value is expected to be binomially distributed. Given a sufficiently large number of trials (N), the p-value is expected to take a gaussian distribution with a mean of the p-value and a variance $\sqrt{p(1-p)/N}$.

The uncertainty is calculated from the first order Taylor series expansion assuming no correlation between bins in the test-statistic distribution and the number of events in each bin follows a Poisson distribution. The calculation of the uncertainty summing over all toy experiments, for the right sided rate-only p-value, is shown as an example and is as follows:

The p-value calculation shown in equation 6 is written in terms of an integral over the test-statistic distribution. This is re-written as a sum of all bins that contribute to the right-tail of the distribution

$$\text{p-value(right tail)} = p = \frac{\int_{N_{obs}}^{+\infty} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} = \frac{\sum_{i=0}^{n(right-tail)} n_i}{\sum_{i=0}^{n(total)} n_i} \tag{11}$$

Where n_i is the the number of times each toy, i, appears. The total uncertainty as a result of the statistical uncertainty on each sample is given by

$$\left(\sigma_p\right)^2 = \sum_i \left(\frac{\partial p}{\partial n_i}\sigma_{n_i}\right) \tag{12}$$

Expanding and using the Poisson uncertainty on the toys $\sigma_{n_i} = \sqrt{n_i}$ before taking the limit where each toy appears only once gives

$$\left(\sigma_p\right)^2 = \frac{p(1-p)}{N_{tot}} \tag{13}$$

Where N_{tot} is the integral of the test-statistic distribution. This is the expected result for a binomial process with a success rate p.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	175.69877	0.00417	11.13658	0.00020	0.02258	0.00015	186.86245
CC1pi	27.64351	0.00201	2.64220	0.00009	0.02418	0.00007	30.31206
CCcoherent	0.28884	0.00000	0.09306	0.00000	0.00000	0.00000	0.38190
2p2h	35.49529	0.00154	1.42670	0.00008	0.02578	0.00004	36.94944
CCother	5.49849	0.00087	0.41118	0.00008	0.00011	0.00001	5.91073
NC1pi0	0.61317	0.01856	0.02142	0.00145	N/A	N/A	0.65460
NC1piPM	4.89912	0.09882	0.17708	0.00990	N/A	N/A	5.18492
NCcoherent	0.00000	0.00000	0.00044	0.00004	N/A	N/A	0.00047
NCother	2.06264	0.07441	0.13205	0.00926	N/A	N/A	2.27837
Total	252.19982	0.20038	16.04070	0.02110	0.07266	0.00027	268.53493

Table 3: Predicted number of events in the neutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of $1.493\,797\,7 \times 10^{21}$ FHC POT. The Asimov data set A in Tab 2 is used. The total of 268.53493 events consists of 252.19982 signal events and 16.33511 background events.

3. Sensitivity to $\bar{\nu}_e$ Appearance

Since there are no fitted parameters in the analysis, the validation can be performed by comparing the rate-only with rate+shape p-values and checking if the corresponding distributions are reasonable, where the final validation is given if all the oscillation analyses agree. This is done by performing a sensitivity study to $\bar{\nu}_e$ appearance data given the Asimov A data set, which is treated as we would treat the data. The Asimov A data set consists of the most probable data set that is the nominal expected E_{reco} vs θ distribution [67] and the corresponding parameters are listed in Tab 2. The expected event rates for all five Super-K event samples using Asimov A parameters are shown in Tables 3 to 7. A further check is performed by comparing the distribution of p-values found for 20k toys data sets.

Parameter(s)	Asimov A
$\sin^2\theta_{23}$	0.528
$\sin^2\theta_{13}$ reactors	0.0219
$\sin^2\theta_{12}$	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601
Mass Hierarchy	Normal
β	1

Table 2: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies corresponding to the Asimov A data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [83], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [22]. The mass hierarchy is not marginalised but fixed to either NH or IH.

The significance to $\bar{\nu}_e$ appearance is quoted by producing p-values using the procedure described in Sec 2. In addition, the mean p-value is also computed by considering the distribution of p-values for the rate-only analysis.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.15943	6.36106	0.00638	0.25944	46.48200	0.28047	53.54879
CC1pi	0.04653	0.93873	0.00282	0.06851	4.48395	0.04775	5.58828
CCcoherent	0.00011	0.00836	0.00018	0.00403	0.03441	0.00330	0.05039
2p2h	0.02985	1.60917	0.00118	0.04339	8.69070	0.03920	10.41351
CCother	0.01337	0.12912	0.00035	0.00957	0.08761	0.00328	0.24329
NC1pi0	1.69779	0.03737	0.06073	0.00362	N/A	N/A	1.79950
NC1piPM	0.16808	0.00476	0.01032	0.00058	N/A	N/A	0.18374
NCcoherent	0.52906	0.00721	0.04837	0.00298	N/A	N/A	0.58763
NCother	0.31903	0.01710	0.02145	0.00130	N/A	N/A	0.35888
NC1gamma	0.94061	0.01701	0.05029	0.00237	N/A	N/A	1.01028
Total	3.90385	9.12991	0.20207	0.39579	59.77868	0.37400	73.78429

Table 4: Predicted number of events in the neutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of $1.493\,797\,7 \times 10^{21}$ FHC POT. The Asimov data set A in Tab 2 is used. The total of 73.78429 events consists of 59.77868 signal events and 14.00562 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	22.71950	0.00126	44.41840	0.00138	0.00140	0.00204	67.14397
CC1pi	5.87728	0.00050	6.74025	0.00037	0.00089	0.00091	12.62020
CCcoherent	0.06995	0.00000	0.26113	0.00007	0.00000	0.00009	0.33125
2p2h	6.64836	0.00038	4.23289	0.00040	0.00065	0.00031	10.88300
CCother	1.40166	0.00018	0.90187	0.00000	0.00000	0.00000	2.30370
NC1pi0	0.08919	0.00330	0.09127	0.00251	N/A	N/A	0.18627
NC1piPM	0.53424	0.02270	0.68366	0.01835	N/A	N/A	1.25895
NCcoherent	0.00000	0.00000	0.00299	0.00000	N/A	N/A	0.00299
NCother	0.46952	0.02440	0.29873	0.01173	N/A	N/A	0.80438
Total	37.80970	0.05271	57.63119	0.03482	0.00294	0.00335	95.53471

Table 5: Predicted number of events in the antineutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of $1.123\,559 \times 10^{21}$ RHC POT. The Asimov data set A in Tab 2 is used. The total of 95.53471 events consists of 57.63119 signal events and 37.90353 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.01912	0.75719	0.03547	1.25946	1.52367	4.23540	7.83030
CC1pi	0.01132	0.15029	0.01089	0.21446	0.21281	0.49728	1.09705
CCcoherent	0.00004	0.00148	0.00062	0.01585	0.00169	0.04650	0.06618
2p2h	0.00437	0.21366	0.00326	0.16792	0.34562	0.49049	1.22533
CCother	0.00578	0.03146	0.00178	0.01716	0.01547	0.00993	0.08158
NC1pi0	0.20578	0.00804	0.26242	0.00695	N/A	N/A	0.48317
NC1piPM	0.03677	0.00189	0.02971	0.00120	N/A	N/A	0.06958
NCcoherent	0.07106	0.00329	0.30096	0.00472	N/A	N/A	0.38002
NCother	0.08422	0.00327	0.03969	0.00147	N/A	N/A	0.12865
NC1gamma	0.11798	0.00653	0.25971	0.00570	N/A	N/A	0.38992
Total	0.55643	1.17710	0.94452	1.69488	2.09927	5.27959	11.75180

Table 6: Predicted number of events in the antineutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of $1.123\,559 \times 10^{21}$ RHC POT. The Asimov data set A in Tab 2 is used. The total of 11.75180 events consists of 5.27959 signal events and 6.47220 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.02845	0.02467	0.00118	0.00177	0.21197	0.00268	0.27071
CC1pi	0.07487	0.76673	0.00258	0.00389	4.85230	0.00229	5.70266
CCcoherent	0.00026	0.01781	0.00008	0.00007	0.11156	0.00011	0.12989
2p2h	0.01000	0.02214	0.00049	0.00096	0.15884	0.00106	0.19349
CCother	0.03763	0.11311	0.00117	0.00284	0.11952	0.00151	0.27577
NC1pi0	0.02039	0.00076	0.00084	0.00008	N/A	N/A	0.02208
NC1piPM	0.06503	0.00166	0.00386	0.00033	N/A	N/A	0.07089
NCother	0.20699	0.00838	0.01319	0.00090	N/A	N/A	0.22945
NC1gamma	0.02853	0.00000	0.00068	0.00007	N/A	N/A	0.02928
Total	0.47215	0.95526	0.02408	0.01091	5.45418	0.00764	6.92422

Table 7: Predicted number of events in the neutrino mode ν_eCC1π^+-like sample obtained after applying the BANFF weight for a total exposure of $1.493\,797\,7 \times 10^{21}$ FHC POT. The Asimov data set A in Tab 2 is used. The total of 6.92422 events consists of 5.45418 signal events and 1.47004 background events.

β	Analysis	p-value	σ	Asimov A Expectation
0	rate-only	0.0237 ± 0.0003	2.262	N.A.
	rate+shape	0.0107 ± 0.0002	2.552	
1	rate-only	0.353 ± 0.001	0.298	$N_{events} = 11.75 \pm 3.43$
	rate+shape	0.343 ± 0.001	0.948	$\Delta\chi^2 = 4.592$

Table 8: Asimov A p-values and event-rates for the $\beta = 0$ and $\beta = 1$ hypotheses using both the rate-only and rate+shape analyses.

The expected Run 1-9c statistics sensitivity studies were performed for $1.493\,797\,7 \times 10^{21}$ POT and $1.123\,559 \times 10^{21}$ POT in neutrino mode and antineutrino mode respectively. The event-rate distributions for both hypotheses are shown in Fig 1 and the $\Delta\chi^2$ distributions for both hypotheses are shown in Fig 2. The corresponding Asimov A p-values are shown in Tab 8.

From the study of the Asimov data set A, which expects approximately 5 signal and 6 background events in the antineutrino mode e-like sample, we expect a p-value of about $1 - 2\,\%$ to exclude the $\beta = 0$ hypothesis, corresponding to about $2.3 - 2.6\,\sigma$, where the conversion to σ is done using equation 14. We find a very large p-value of about $30\,\%$ in both analyses to exclude the $\beta = 1$ hypothesis, corresponding to $< 1\,\sigma$, which is expected as Asimov data set A was generated using the $\beta = 1$ hypothesis.

$$\sigma = \sqrt{2}\,\mathrm{erf}^{-1}\,(1 - p) \tag{14}$$

[NOTE: All figures and results shown for the sensitivity are currently from the un-weighted analysis, as I am waiting for the weighted analysis results]

[Insert weighted p-value distributions]

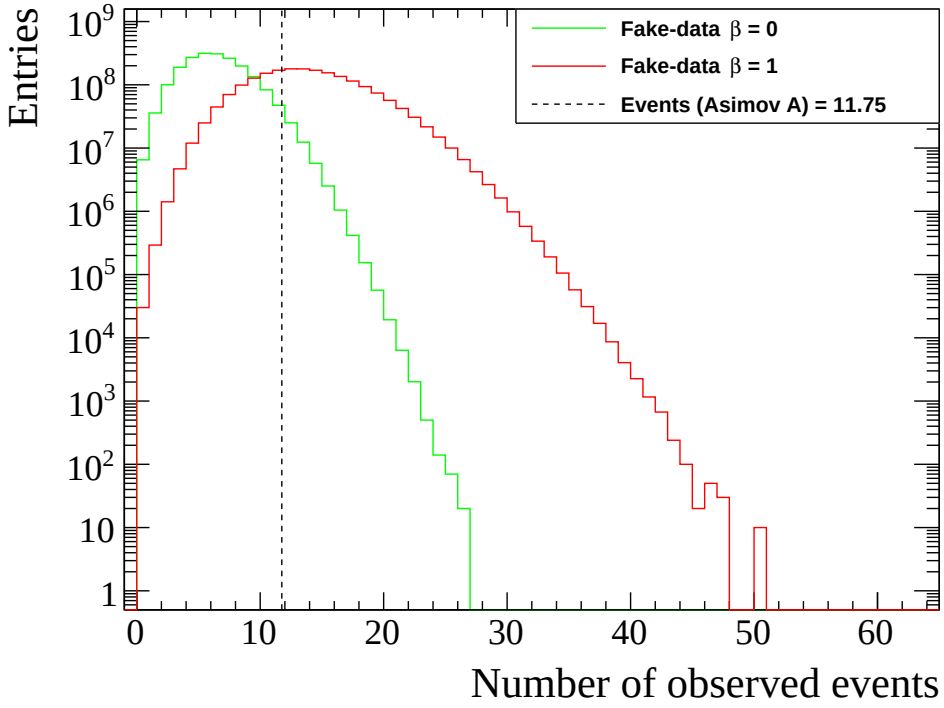


Figure 1: The distribution of the total number of events from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The dashed line corresponds to the total number of events given by the Asimov A data set.

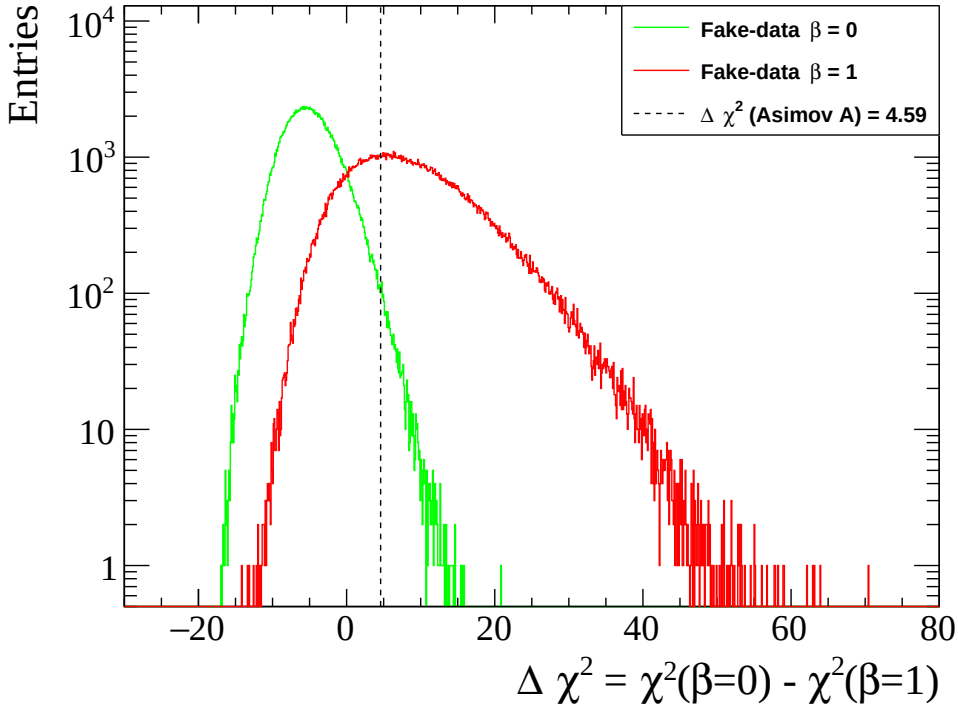


Figure 2: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The dashed line corresponds to the $\Delta\chi^2$ value given by the Asimov A data set.

	Rate-only Analysis			Rate+shape Analysis			
	N_{obs}	$\beta = 0$ p-value	$\beta = 1$ p-value	$\Delta\chi^2$	$\beta = 0$ p-value	$\beta = 1$ p-value	B_{01}
Run 1-9c		$\pm$	$\pm$		$\pm$	$\pm$	
Run 1-7	4	0.410	0.2155	-1.930	0.3742	0.099	2.62

Table 9: Summary of observed number of events, $\Delta\chi^2$, rate-only and rate+shape p-values and Bayes factor from the combined Run 1-9c data set compared to the Run 1-7 data set used for the previous iteration of this analysis, in 2016.

4. Results of the $\bar{\nu}_e$ Appearance Analysis

The rate-only and rate+shape p-values for $\bar{\nu}_e$ appearance are performed on the combined Run 1-9c statistics following the same procedure as in Sec 2. 1-sided p-values are computed on the right tail of the test statistic expected distribution to exclude $\beta = 0$ and the left tail to exclude $\beta = 1$. Fig **??** shows the data in the 5 Super-K event samples overlaid with the expected event rates for Asimov data set A. A summary of the observed number of events in Super-K, $\Delta\chi^2$ and p-values is given in Tab 9.

In Fig **??** the distributions of the test statistic, for rate-only and rate+shape analyses, are shown and compared with the data. Tab 9 contains a summary of the results. [Insert statement about how much the data disfavours each hypothesis - Will run with data after the weighted sensitivity has been run]

4.1. Comparison to the Results of the 2016 (Run 1-7) Analysis

A comparison of the Run 1-9c and Run 1-7 analysis results are shown in Tab 9. The Run 1-9c data set prefers the $\beta =$[hypothesis] hypothesis. [statement about which analyses prefer which hypotheses - Will run with data after the weighted sensitivity has been run]

5. Summary

In this technical note, we have presented the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour $\nu/\bar{\nu}$ joint analysis framework performed by the VALOR group, on the combined Run 1-9c data set, for an exposure of $1.493\,797\,7 \times 10^{21}$ POT in neutrino mode and $1.123\,559 \times 10^{21}$ POT in antineutrino mode. The goal of the analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam.

This analysis predicts 268.535 ± 16.387 (syst) for neutrino μ-like, 95.5347 ± 9.7742 (syst) for antineutrino μ-like, 73.7843 ± 8.5898 (syst) for neutrino e-like, 11.75 ± 3.43 (syst) for antineutrino e-like and 6.92422 ± 2.63239 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively for neutrino μ-like, for antineutrino μ-like, for neutrino e-like, for antineutrino e-like and for neutrino ν_eCC1π^+-like events.

Both a rate only and a rate+shape analysis are performed. In both analyses, fits were performed in the joint fit framework, including all samples as data to constraint either background and the hypotheses tested. The number of observed events in the antineutrino mode e-like spectrum is and the measured rate+shape test statistic, $\Delta\chi^2 = \chi^2\,(\beta = 0) - \chi^2\,(\beta = 1)$ was found to be . The significance of the result is shown as a p-value. Taking no $\bar{\nu}_e$ appearance as our null hypothesis we

find a p-value of ±for the rate-only analysis and a p-value of ±for the rate+shape analysis. Taking $\bar{\nu}_e$ appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of ±for the rate-only analysis and a p-value of ±for the rate+shape analysis.

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Appendices